

Finance course

Session 14 – 12/02/2026



Value



ALBERTSCHOOL

Introduction

Semester 1: Introduction to Financial Markets

- 1.1 Presentation of the financial markets (4h)
- 1.2 Mechanisms of financial markets (2h)
- 1.3 Risk in finance (2h)
- 1.4 Equities (2h)
- 1.5 Bonds (2h)
- 1.6 Options (2h)
- 1.7 Intervention (2h)
- 1.8 Financial market regulation (2h)
- 1.9 Case study (2h)

Semester 2 : Value

- 2.1 Reminder of accounting principles (2h)
- 2.2 Financial Analysis (2h)
- 2.3 Cost of capital (2h)
- 2.4 Financial structure (2h)
- 2.5 Intervention
- 2.6 Methods of valuing a business (6h)
- 2.7 Valuation of a company (4h)
- 2.8 Capital transactions (6h)

Understanding the value of a company compared to its risk

Press Review

Educational links:

Financial Analysis :

<https://www.youtube.com/watch?v=eXLbXZSonoM>

Documentary :

Banking on Bitcoin (2016): <https://www.youtube.com/watch?v=8rsxIW02W0g>

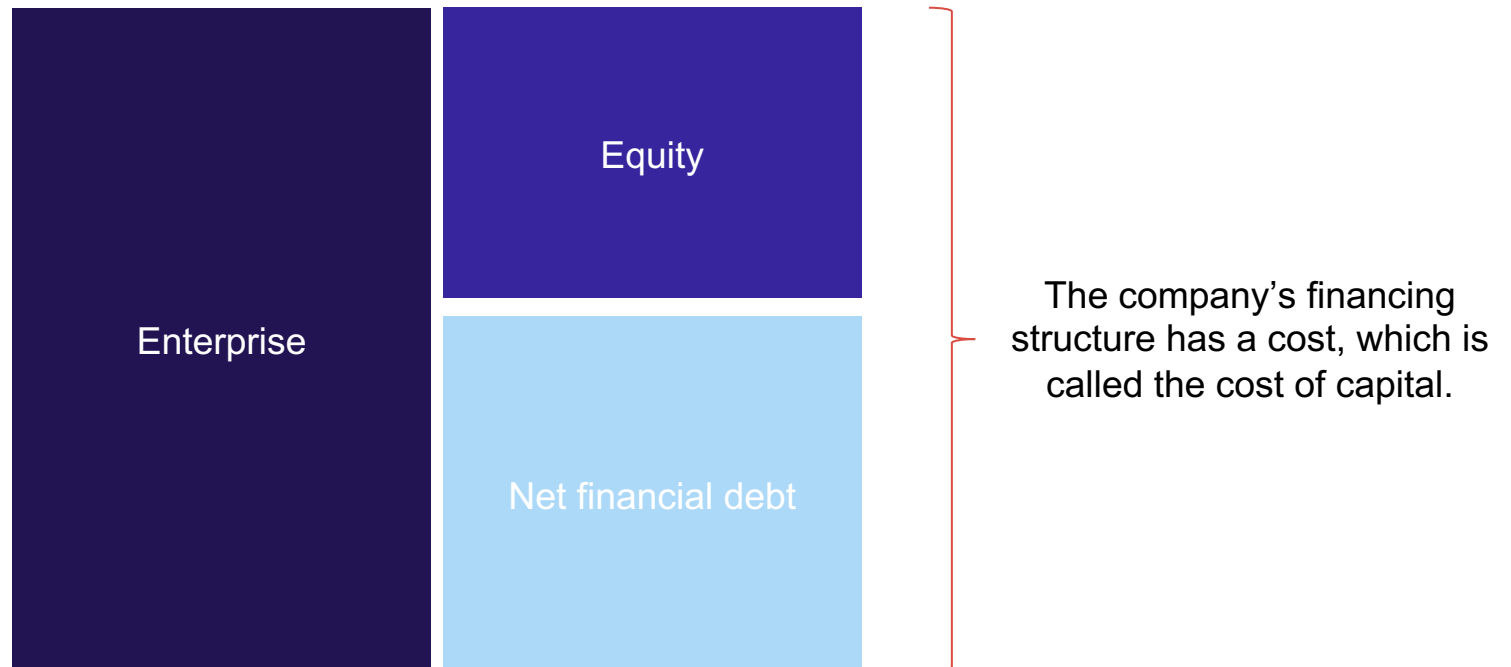
What is WACC?

The cost of capital is the minimum rate of return that a company's investments must generate in order to satisfy both:
the return required by shareholders (i.e. the cost of equity), and
the return required by creditors (i.e. the cost of debt).

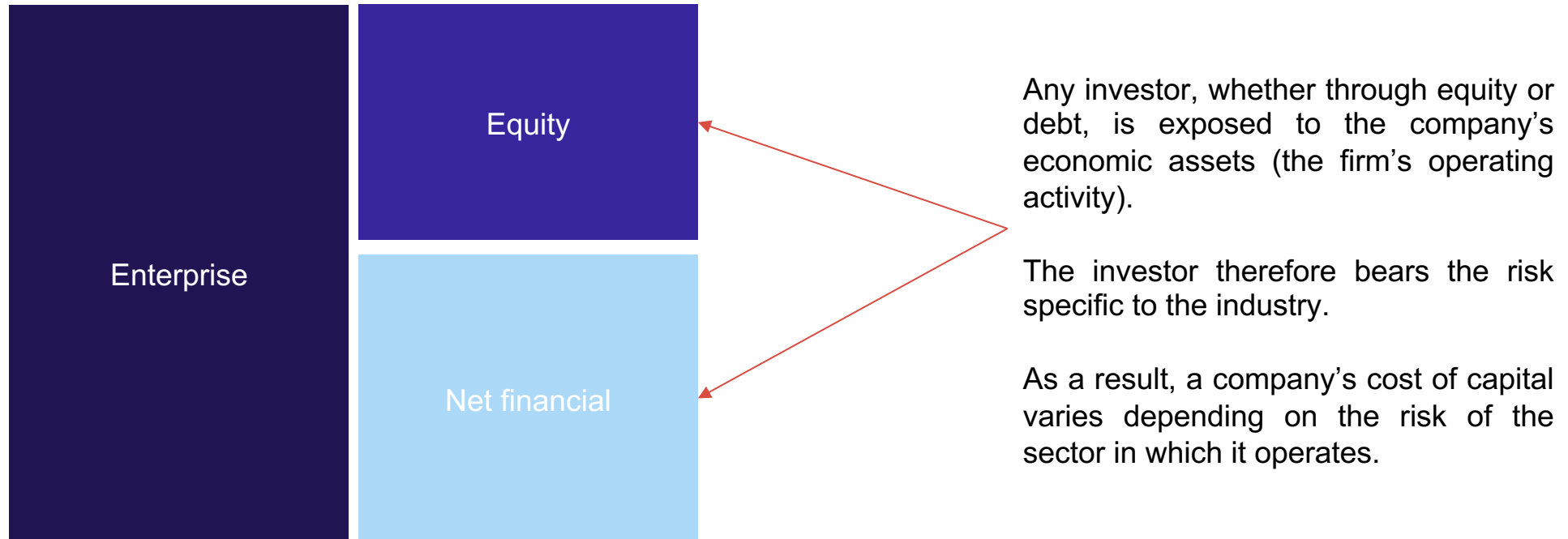
⇒ The cost of capital is therefore the company's overall cost of financing.

Warning: do not confuse the cost of capital with the cost of equity.

What is WACC?



What is WACC?



Economic characteristics of an industry

Sensitivity to the economic cycle

Some sectors are more sensitive to the economic cycle, such as transportation and construction (high cost of capital), while others are less sensitive, such as agri-food or healthcare (low cost of capital).

Cost structure

Some sectors are mainly based on **variable costs**, while others rely heavily on **fixed costs**. A sector with high fixed costs has a higher cost of capital because it is more sensitive to economic fluctuations.

Business visibility

The more predictable the activity (long-term contracts), the lower the cost of capital.

Growth rate of earnings

The higher the growth rate, the higher the cost of capital, because most of the company's value is based on its future results.

Calculating the WACC

2 methods

1. Direct method using unlevered Beta
2. Indirect Method

Calculating the WACC

Direct method using the β of economic assets

The rate of return required for any investment to satisfy an investor is equal to:

- the **risk-free rate**, plus
- a **risk premium** that compensates for the risk related to the company's specific activity.

$$R_A = R_f + \beta_A \times (R_m - R_f)$$

R_A : Coût du capital économique (hors effet de levier)

R_f : Taux sans risque (ex. : taux des obligations d'État)

R_m : Rendement attendu du marché (ex. : rendement moyen d'un indice)

$R_m - R_f$: Prime de risque du marché

β_A : Bêta de l'actif économique (désendetté)

Calculating the WACC

The β of economic assets

Just as β measures the sensitivity of a financial security relative to the market, the β **of an economic asset** measures the dispersion of the future cash flows of that asset relative to those of the market.

These two β s are interdependent: a company that invests in high-risk projects (high β) will have a high equity β .

Bêta de l'actif économique		Bêta de l'actif économique	
		Tous secteurs	0,89
Distribution alimentaire	0,48	Hôtellerie	0,99
Autoroutes	0,51	Production d'électricité	1,00
Biens de consommation	0,61	Logiciels	1,04
Agroalimentaire	0,61	Équipementiers automobiles	1,06
Bière	0,63	Travail temporaire	1,08
Tabac	0,65	Services informatiques	1,08
Télécom	0,65	Machine outils	1,11
Média / publicité	0,72	Technologies	1,12
Distribution pharmaceutique	0,75	Assurance vie	1,16
Distribution d'électricité	0,81	Production d'électricité éolienne	1,30
Distribution automobile	0,90	Services pétroliers	1,40
Matériaux de construction	0,93	Mines	1,73

Source BNP Paribas, Corporate Finance, équipe BVT

Calculating the WACC

The β of economic assets

The β of economic assets can be reconstructed, as it is equal to the **weighted average of the equity β and the net debt β** .

$$\beta_A = \frac{E}{E + D(1 - T)} \times \beta_E + \frac{D(1 - T)}{E + D(1 - T)} \times \beta_D$$

E : Valeur des fonds propres (Equity)

D : Valeur de la dette

T : Taux d'imposition

β_E : Bêta des fonds propres

β_D : Bêta de la dette (souvent proche de zéro)

R_E : Coût des fonds propres

R_D : Coût de la dette

Considering the debt is risk-free, $\beta_D = 0$

$$\beta_{\text{actif économique}} = \frac{\beta_{\text{capitaux propres}}}{1 + (1 - IS) \times \frac{V_d}{V_{cp}}}$$

Calculating the WACC

Indirect method

Since we are looking for the return required by all providers of funds to the company, it is possible to reconstruct the cost of capital by valuing the different financial securities issued by the firm.

We use the **Weighted Average Cost of Capital (WACC)**.

$$WACC = \frac{E}{E+D} \times R_E + \frac{D}{E+D} \times R_D \times (1 - T)$$

E : Valeur des fonds propres (Equity)

D : Valeur de la dette

T : Taux d'imposition

R_E : Coût des fonds propres

R_D : Coût de la dette

Calculating the WACC

Indirect method – Example

A company financed by:

- 100 of equity with a cost of 8%, and
- 50 of debt with a cost of 3% before tax at 25%,
has a cost of capital of:

$$WACC = \frac{E}{E+D} \times R_E + \frac{D}{E+D} \times R_D \times (1 - T)$$

Calculating the WACC

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$$WACC = \frac{E}{E+D} \times R_E + \frac{D}{E+D} \times R_D \times (1 - T)$$

$$WACC = 8\% \times \frac{100}{100 + 50} + 3\% \times (1 - 25\%) \times \frac{50}{100 + 50}$$

$$WACC = 6,1\%$$

Calculating the WACC

Indirect method – Question

If the cost of debt is lower than the cost of equity, why not maximize the share of financial debt in order to reduce the WACC?

Calculating the WACC

Indirect method – Question

If the cost of debt is lower than the cost of equity, why not maximize the share of financial debt in order to reduce the WACC?

- The beta increases, which increases the cost of equity.
- The WACC therefore remains unchanged.

⇒ **Calculation of the cost of equity**

$$R_E = R_f + \beta_E \times (R_m - R_f)$$

R_E : Coût des capitaux propres (taux de rentabilité exigé par les actionnaires)

R_f : Taux sans risque (ex. : obligations d'État)

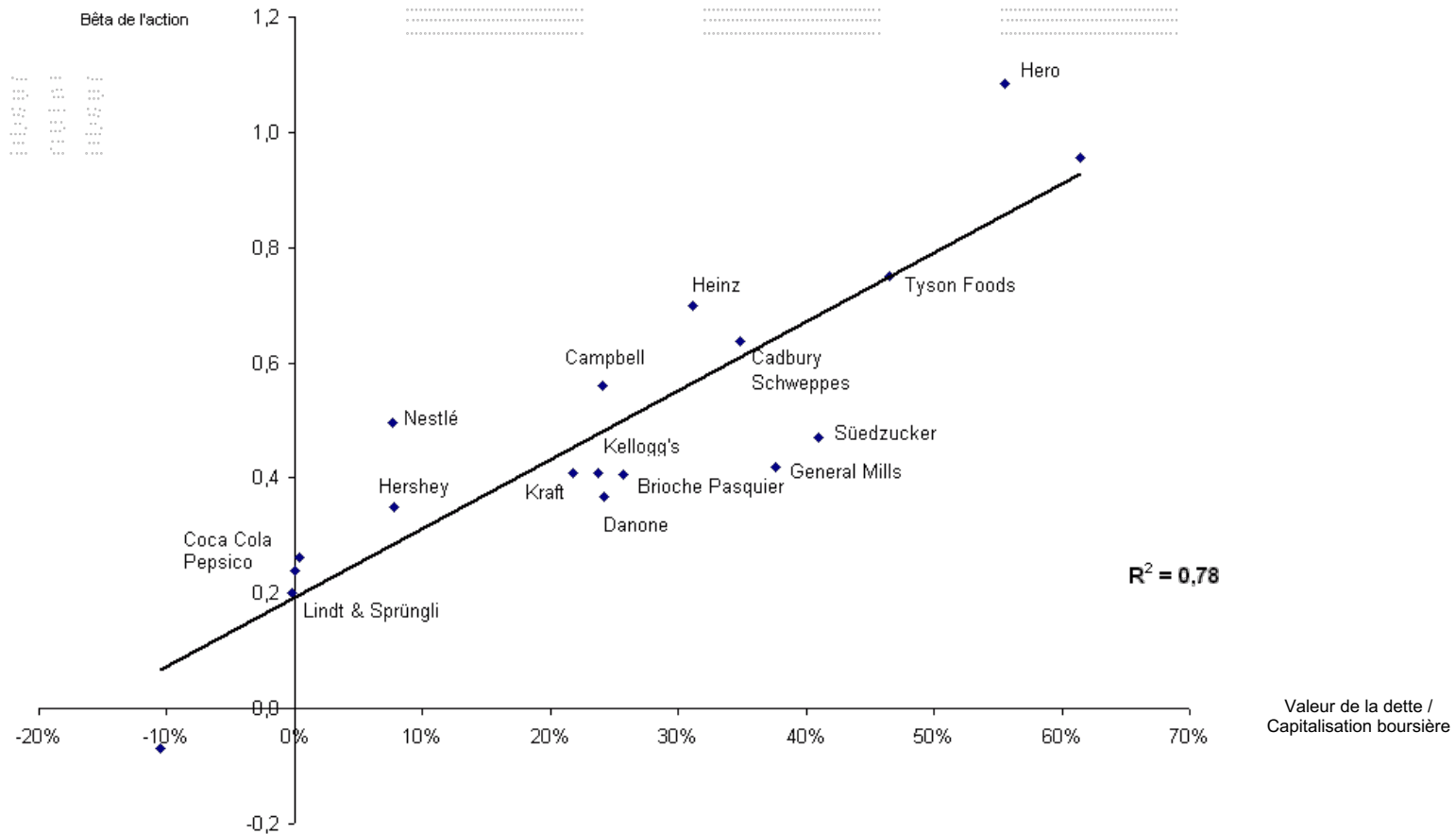
β_E : Bêta des fonds propres (mesure de la sensibilité du titre au marché)

R_m : Rendement attendu du marché

$R_m - R_f$: Prime de risque du marché

Calculating the WACC

influence of the level of leverage on the equity beta of firms in the agri-food sector.



Exercise

Partners wish to start a real estate development company. They are considering the financial structure and are hesitating between three different capital structures:

Leverage ratio %	Cost of Equity %
0	8
50	12
90	17

- Cost of debt before tax: **6%**
- Corporate tax rate: **25%**

- 1/ Calculate the WACC for the three different capital structures.**
- 2/ Which one would you recommend? What are the limitations?**